Practice workbook · Explain and evaluate uses of embryonic and adult stem cells

Why unspecialised stem cells are useful, treating disease by differentiation, and weighing benefits, risks and ethics of embryonic, adult and induced cells.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EVALUATE-STEM-CELL-USE

1. Question 8 · part whole · 6 marks

June 2022 Paper 1

Worked-example retry · Attempt part whole only.

8 Discuss the advantages and disadvantages of the use of embryonic and adult stem cells in medical research.

(6)

embryonic stem cells

adult stem cells

(Total for Question 8 = 6 marks)

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1. Question 8 · part whole · 6 marks · Working and answer

Working and answer

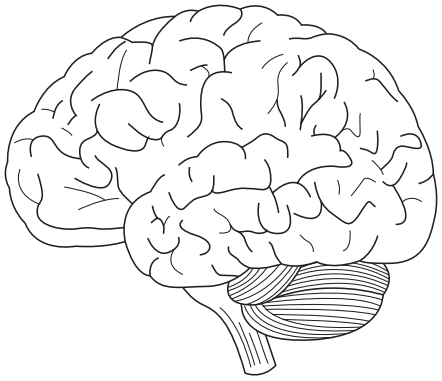
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2. Question 7 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

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20

2. Question 7 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

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(iii) Explain how one neurone communicates with another neurone.

(3)

(c) Recent scientific research into the treatment of Parkinson's disease has involved the use of stem cells.

Explain how stem cells could be used to reduce the symptoms of Parkinson's disease.

(3)

(Total for Question 7 = 11 marks)

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2. Question 7 · part c · 3 marks · Working and answer

Working and answer

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3. Question 6 · part c · 1 mark

June 2023 Paper 2R

Further practice · Attempt part c only.

6 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Pregnancy begins with the formation of a zygote. This stage is known as conception. The zygote develops into an embryo. By day five the embryo is transformed into a blastocyst, a hollow ball of about 100 genetically identical cells. These cells have not yet differentiated and they can provide a source of embryonic stem cells for use in medical research. Embryos produced following IVF treatment can be used as a source of embryonic stem cells.

In the second week following conception, the blastocyst begins the process of implantation and releases a hormone, called hCG. This hormone maintains the corpus luteum, a structure formed in the ovary following ovulation. After implantation the amniotic sac forms, which is where the blastocyst and later the fetus develop.

In the fourth and fifth week after conception, the umbilical cord forms. This connects the embryo to the placenta by week eight. The placenta is not fully developed until week ten when it begins to produce progesterone and oestrogen. All organ systems are present at this time in pregnancy.

Fetal development begins at about month four and it is possible to distinguish males from females at this time. In males the testes, which develop in the abdominal cavity, release testosterone. Female characteristics are produced in the fetus if testosterone is not released at the stage.

Hormones are important in starting labour and the birth process, about nine months after conception. The cervix dilates, with the head of the baby pushing against the cervix to help the process of dilation. Once the child is born and breathing normally, the umbilical cord is tied and cut, separating the child from the placenta.

(a) Describe the process that occurs at conception. (lines 1 and 2)

(2)

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3. Question 6 · part c · 1 mark

June 2023 Paper 2R

Further practice · Attempt part c only.

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(b) Describe the process that forms a blastocyst. (lines 2 to 4)

(4)

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(c) Which is a reason why cells taken from a blastocyst can be useful in scientific research?

(1)

- ☒ **A** the cells are already differentiated
- ☒ **B** the cells are able to differentiate into both sex cells and body cells
- ☒ **C** the cells are able to differentiate into any type of sex cell
- ☒ **D** the cells are able to differentiate into any type of body cell

3. Question 6 · part c · 1 mark

June 2023 Paper 2R

Further practice · Attempt part c only.

(d) Explain why it is important that hCG maintains the corpus luteum. (lines 7 to 9)

(2)

(e) Explain the importance of the development that takes place in the fourth and fifth week after conception. (lines 12 and 13)

(2)

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(f) Suggest the role of testosterone in the development of the fetus. (lines 17 to 19)

(1)

(g) Which change occurs during the birth process? (line 20)

(1)

- ☒ **A** oestrogen levels decrease
- ☒ **B** progesterone levels increase
- ☒ **C** release of oxytocin from the pituitary gland
- ☒ **D** release of hCG from the placenta

(Total for Question 6 = 13 marks)

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3. Question 6 · part c · 1 mark · Working and answer

Working and answer

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4. Question 6 · part a.i · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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4. Question 6 · part a.i · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

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2 .....

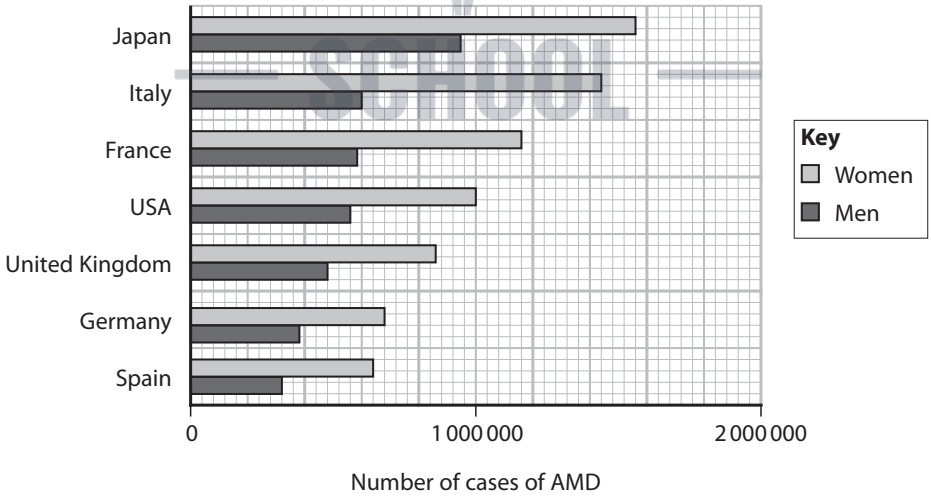
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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4. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

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5. Question 6 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(3)

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5. Question 6 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

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2 .....

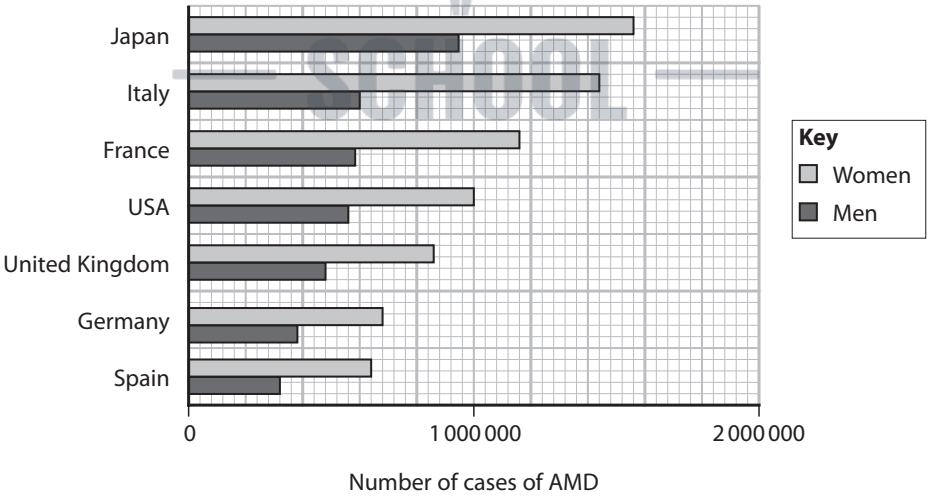
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Turn over ▶

5. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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